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COP 4834 Web Systems II

Assignment 11: Application Testing

Professor Dr. Ron Eaglin

Assignment:

For this assignment we will develop a test plan for the software in the application. We will include multiple unit tests and end user testing. Each interface should be listed along with the associated test plan.

Title: Application Testing

Objectives:

Develop a software test plan

Project Overview:

The test plan for this application is to test a CRUD web application that can allow the users to create, view, update, and delete user accounts. The purpose of this test plan is to make sure that all the main functions of the application work properly.

Testing Objectives:

The objective for testing this application is to confirm that all of the applications functions work correctly and can handle any valid or invalid data we put in it.

Features being tested:

Some of the features being tested are listed below.

Interface/Feature	Testing	Type of test
Login Page	Valid and invalid log in attempts	Manual/ Integration
Create User form	Adding a new user	Manual/ Unit
View Users page	Displaying all the users from the database	Integration
Update user button	Editing any existing user information	Manual/ Integration
Delete user button	Remove or hide user	Manual/ Integration

Admin Accessibility	Admin can add/update/delete	Manual/ Integration
Database connection	Saves data and can load correctly	Integration

Testing tools:

The tools we will be using to do the unit tests will be in replit where the application is currently working and any needed updates can be made inside.

Test Case 1: Valid Login

Field	Description
Test Case ID	T-001
Feature	Login
Test Type	Manual/ Integration
Input	Email: admin@test.com Password: admin123
Steps	Open login page and enter admin credentials then click login
Expected Result	User is logged in and admin options appear
Actual Result	Login was successful and admin options appeared
Pass/Fail	Pass

Test Case 2: Invalid Login

Field	Description
Test Case ID	T-002

Feature	Login
Test Type	Manual/ Integration
Input	In admin login Email: user@test.com Password: user123
Steps	Enter invalid credentials and click login
Expected Result	Error message pops up saying “Login Failed”
Actual Result	An error message popped up in the bottom right corner saying “ Login Failed”
Pass/Fail	Pass

Test Case 3: Create User

Field	Description
Test Case ID	T-003
Feature	Create
Test Type	Manual/ Integration
Input	Username: testuser Email: testuser@email.com Password: test123 Role: Admin
Steps	Fill out the text box fields
Expected Result	New User appears in the registered user list
Actual Result	New user was added and visible with the other registered users

Pass/Fail	Pass
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Test Case 4: View Users

Field	Description
Test Case ID	T-004
Feature	Read
Test Type	Integration
Input	No input needed/open users page
Steps	Load the application homepage
Expected Result	Any existing users should be available on the homepage
Actual Result	There are 2 existing users displayed
Pass/Fail	Pass

Test Case 5: Update User

Field	Description
Test Case ID	T-005
Feature	Update
Test Type	Manual/ Integration
Input	Changed information (username or email)
Steps	Click edit changes and enter the new user information then save the changes

Expected Result	After refreshing the page the new changes should be visible when the user tries to log in
Actual Result	The new changes applied were successful
Pass/Fail	Pass

Test Case 6: Delete User

Field	Description
Test Case ID	T-006
Feature	Delete
Test Type	Manual/ Integration
Input	No input needed
Steps	Log in as admin and then choose a user and click the delete button
Expected Result	User is removed or hidden from the registered user list
Actual Result	The chosen user is now gone from the registered user list
Pass/Fail	Pass

Test Case 7: Role-Based Access

Field	Description
Test Case ID	T-007
Feature	Security/Role Management
Test Type	Manual/ Integration

Input	User credentials at the login page
Steps	Login under regular user and try to access the update or delete option
Expected Result	Regular user cannot update or delete any user records
Actual Result	Regular user was not able to view any of the edit user buttons
Pass/Fail	Pass

Integration Testing:

The integration testing will help verify that the user interface, servers, and database can all work together cohesively.

End User Testing:

We will perform end user testing by using the application in a browser as a normal user. The testee will then create a user, view them on the user list, and update the user information, then delete the user. The results are recorded above in the test case tables.

Conclusion:

This test plan helps provide a structured method to testing the main parts of the application to make sure they are functioning correctly. This plan includes unit tests for each of the functions of the application including integration tests, and end user tests.

Appendix A: Test Case Screenshots

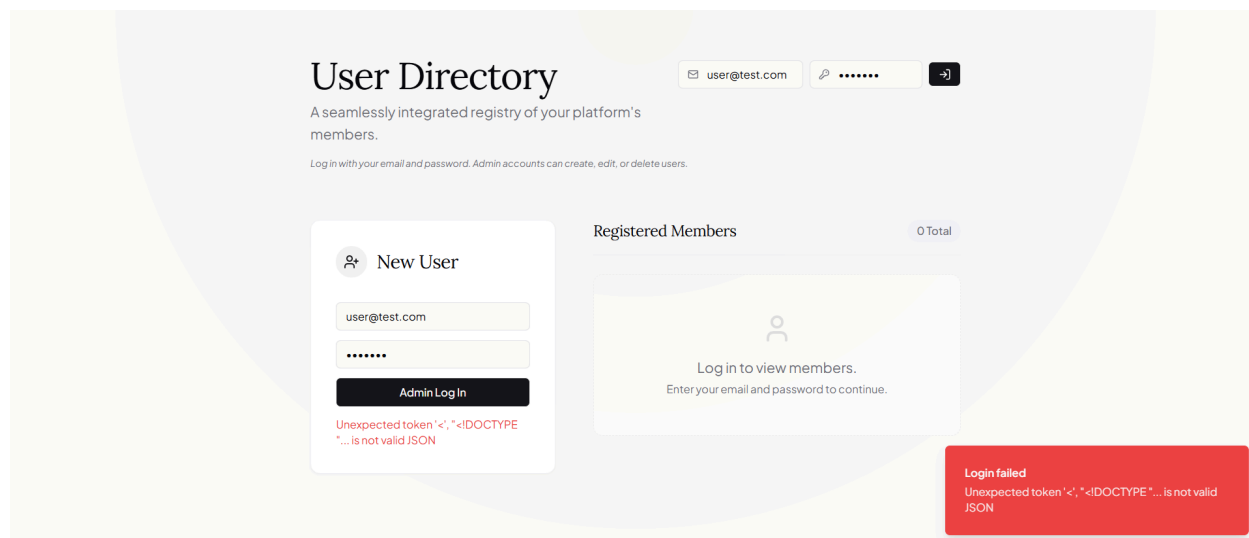
Valid Login: Admin

The screenshot shows the 'User Directory' interface. At the top right, the user is logged in as 'admin' with a red 'admin' label and an edit icon. The main heading is 'User Directory' with the subtitle 'A seamlessly integrated registry of your platform's members.' On the left, there is a 'New User' form with fields for 'USERNAME' (containing 'johndoe'), 'EMAIL ADDRESS' (containing 'john@example.com'), 'PASSWORD' (with a 'Min. 6 characters' hint), and a 'ROLE' dropdown menu set to 'Select a role'. A 'Register User' button is at the bottom of the form. On the right, the 'Registered Members' section shows '0 Total' and a message: 'No users registered yet. Use the form to create your first member.'

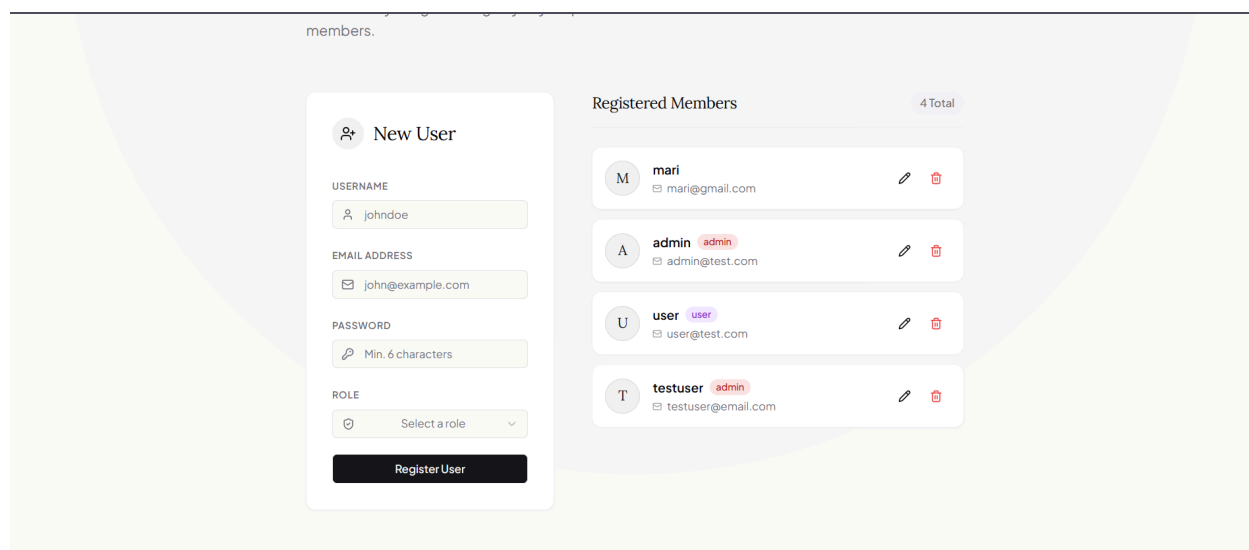
Valid Login: Regular User/ Role Based Management

The screenshot shows the 'User Directory' interface as a regular user. At the top right, the user is logged in as 'user' with a purple 'user' label and an edit icon. The main heading is 'User Directory' with the subtitle 'A seamlessly integrated registry of your platform's members.' Below the subtitle, a warning message reads: 'Viewing as user — admin access required to make changes.' On the left, there is an 'Admin Log In' form with fields for 'Admin email' and 'Admin password', and an 'Admin Log In' button. On the right, the 'Registered Members' section shows '0 Total' and the same message: 'No users registered yet. Use the form to create your first member.'

Invalid User



Create User:



Update User:

members.

 New User

USERNAME

 johndoe

EMAIL ADDRESS

 joh@example.com

PASSWORD

 Min. 6 characters

ROLE

 Select a role

Register User

Registered Members

4 Total

M

mari

A

admin admin

U

user user

T

testuser1 admin

Delete User:

members.

 New User

USERNAME

 johndoe

EMAIL ADDRESS

 joh@example.com

PASSWORD

 Min. 6 characters

ROLE

 Select a role

Register User

Registered Members

3 Total

M

mari

A

admin admin

U

user user

Removed

testuser1 has been removed.